

Chapter 5 --- Installing MySQL Server and MySQL Workbench

To practise SQL on your computer, you need to install two applications:

Application Role

MySQL Server Stores databases and executes SQL queries

MySQL Workbench Provides a graphical interface for writing queries and working with MySQL Server

MySQL Workbench alone is not enough --- the server must also be installed and running.

5.1 System Requirements

- A Windows computer
- Administrator access
- A stable internet connection
- At least 2 GB of free storage

5.2 Step-by-Step Installation

Step 1 --- Download the MySQL Installer

- Open the official MySQL Installer download page.
- You will see two installer options --- Web Installer (smaller, downloads components during installation) or Full Installer (larger, contains all required components). Download the Web Installer if you have a stable internet connection.
- Click Download.
- If the website asks you to sign in, click "No thanks, just start my download".

Step 2 --- Start the Installer

- Open the downloaded .msi file and allow it to make changes to your computer.
- Select Developer Default as the setup type. This normally installs MySQL Server, MySQL Workbench, MySQL Shell, and required connectors/tools.
- Click Next. (If Developer Default is unavailable, select Custom and manually choose MySQL Server and MySQL Workbench.)

Step 3 --- Install the Required Components

- The installer may check for additional software required by MySQL
- click Execute to install the requirements.
- Wait until all required components show a completed status, then click Next.
- Click Execute again to install MySQL Server and MySQL Workbench.
- After all products are installed successfully, click Next.

Step 4 --- Configure MySQL Server

Use the following settings and keep everything else at its default:

Setting Value

Server Configuration Standalone MySQL Server

Configuration Type Development Computer

Connectivity TCP/IP

Port 3306

Authentication Strong Password Encryption

Step 5 --- Create the Root Password

The root user is the administrator account for MySQL.

- Enter a password for the root user.
- Re-enter the same password to confirm it.
- Write down or safely store the password --- you will need it to connect through Workbench (Username: root, Password: the one you just created).

Do not forget the root password.

Step 6 --- Configure the Windows Service

MySQL runs as a Windows service in the background.

- Keep "Configure MySQL Server as a Windows Service" selected.
- Keep the default service name shown by the installer.
- Select "Start the MySQL Server at System Startup".
- Keep "Standard System Account" selected, then click Next.

Starting MySQL with Windows ensures the server is available whenever you open Workbench.

Step 7 --- Apply the Configuration

- Click Execute and wait while the installer applies the configuration.
- Confirm that every configuration step shows a success status.

- Click Finish, then continue through the remaining screens to complete the installation.

Step 8 --- Open MySQL Workbench

- Open the Windows Start menu and search for MySQL Workbench.
- Under MySQL Connections, select the local connection (it may appear as "Local instance MySQL" or "Local instance 3306").
- Enter the root password created during installation.
- Optionally select "Save password in vault" so you don't enter it every time, then click OK.

The SQL Editor will open once the connection is successful.

Step 9 --- Verify the Installation

Open a new SQL tab and execute:

```
SELECT VERSION();
```

This displays the installed MySQL Server version. Next, execute:

```
SHOW DATABASES;
```

You should see the databases currently available on your server.

Step 10 --- Create a Test Database

```
CREATE DATABASE sql_course;USE sql_course;CREATE TABLE Students ( StudentID INT PRIMARY KEY, StudentName
```

Expected output:

```
StudentID StudentName
```

```
-----
```

```
101 Aarav
```

If this result appears, MySQL Server and MySQL Workbench have been installed and connected successfully.

5.3 How the Connection Works

You write a query in MySQL Workbench → Workbench sends the query to MySQL Server → MySQL Server executes the query → Workbench displays the result. MySQL Workbench is the interface, while MySQL Server performs the actual database operations.

5.4 Common Installation Problems

Problem Likely Cause / Fix

Cannot connect to MySQL Server The MySQL service may not be running. Press Windows + R, enter services.msc, find the MySQL service, right-click it and select

Start or Restart.

Access denied for user root The entered root password is incorrect. Enter the same password created during server configuration.

Workbench installed but no local MySQL Server may not have been connection appears installed. Run MySQL Installer again and add MySQL Server.

Port 3306 is unavailable Another application may already be using port 3306. Select another available port, such as 3307, and use it when creating the Workbench connection.

Server Connection Settings (Local Installation)

Setting Value

Hostname localhost

Port 3306

Username root

Password Root password created during installation

■ Final CheckYour installation is complete when: MySQL Server is installed, the MySQL Windows service is running, MySQL Workbench opens successfully, Workbench connects to the local server, and SQL queries execute without connection errors.

Interview Questions — Chapter 5

Q1. Is MySQL Workbench the database server?

Ideal answer: No. Workbench is a graphical client/interface; MySQL Server stores data and executes SQL.

Q2. What is the default MySQL port used in the notes?

Ideal answer: 3306.

Q3. What does localhost mean in a local MySQL connection?

Ideal answer: It refers to the same computer on which the MySQL Server is running.

Q4. Why can Workbench open but queries still fail?

Ideal answer: Workbench can be installed while the MySQL Server service is stopped or misconfigured; the client still needs a working server connection.

Q5. How do you verify a successful installation?

Ideal answer: Connect to the server and run `SELECT VERSION()` and `SHOW DATABASES`, then create a small test database/table and execute a `SELECT`.